BullSO 2026 Invitational

Rocks and Minerals B/C

Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Number: _____

Team Name:_____

Participant Names:_____

Total Score:_____/305 **Rank:_____**

General Direction: Points won't be lost from incorrect answers, so you should attempt every question. Using the label (i.e. A, B, C) is fine when referring to a specimen in an answer. Certain questions are labelled "Div C Only," please do not attempt them if you are Div B (middle school). Finally, Div B teams will get +10 points for writing their team number on every page of the answer sheet, and Div C teams will get +1. It must be on all pages. Good Luck!

Station 1

Letter	ID	I/E	F/I/M	High/Med/Low	High/Med/Low
A	Basalt	E	M	Low	High
B	Andesite	E	I	Med	Medium
C	Rhyolite	E	F	High	Low

1. Andesite- Intermediate composition typical of convergent boundaries.

2. Rhyolite < Andseite < Basalt (C<B<A)

3. Basalt- Olivine forms at highest temperatures.

4. Higher Fe/Mg content, lower silica.

Station 2

Letter	ID	Classification	Environment
A	Shale	Clastic	Deep marine/lake
B	Conglomerate	Clastic	River/alluvial fan

1. Conglomerate B- shale is made of fine-grained mud and clay- low energy, while conglomerate needs high energy water to transport & deposit the large clasts that make it up

2. Conglomerates B feature well-rounded to subrounded, gravel-sized clasts >2mm indicating high-energy transport (rivers, beaches) that abrades particles. Sorting varies from poor to moderate. Shale consists of extremely fine, angular clay-sized particles <4μm deposited in low-energy, quiet environments (deep ocean, lakes), generally appearing well-sorted due to uniform, tiny grain size.

3. Shale A- Finer grains, longer transport.

4. Shale A- Plane bedding/laminations
   Conglomerate - Graded bedding/imbrication

Station 3

Letter	ID	F/NF	Protolith	L/M/H grade
A	Quartzite	NF	Sandstone	M
B	Gneiss	F	Granite/Shale	H
C	Marble	NF	Limestone	M
D	Phyllite	F	Shale	L-M

1. Gneiss B - Mineral segregation during high-grade metamorphism.

2. Marble C

Station 4

1. Bituminous Coal
2. Compacted plant material from peat swamps
3. Peat < Lignite < Bituminous < Anthracite
4. Temperature is considered the dominant factor, acting as a catalyst for chemical changes, while pressure contributes to physical compaction, both working together to increase carbon content, density, and calorific value while decreasing moisture and volatile matter
5. Electricity generation, steel production, industrial steam production, smelting iron, manufacturing cement, and powering industrial plants
6. CO_2 emissions / habitat destruction / acid mine drainage

7. Swamp

Station 5

1. Alabaster
2. Sulfate
3. 1.5-2
4. $CaSO_4$ $2H_2O$
5. Evaporating shallow seas / lakes
6. Requires evaporation > precipitation
7. Halite
8. Spain
9. Sculpting, lighting, decoration, sculpture, architecture, fragrance diffusion

Station 6

Letter	ID	M/NM	Streak
A	Pyrite	M	Greenish-black
B	Hematite	M	Red-brown
C	Graphite	NM	Gray-black

1. Pyrite A- brassy color, looks like gold
2. Quartz, calcite, micas and tourmaline (only need 2)
3. Graphite C - delocalized electrons

Station 7

Letter	ID	Formula	Mohs (**accept any in range**)
A	Calcite	$CaCO_3$	3
B	Fossiliferous Limestone	$CaCO_3$	3-4

1. Produces calcium ions, water, and carbon dioxide gas, resulting in a characteristic, vigorous bubbling or effervescence
2. Double refraction
3. Living organisms in marine environments extract calcium carbonate from water to build shells and skeletons, which subsequently accumulate and harden
4. Fossiliferous limestone B

Station 8

Letter	ID	Group	Luster	Specific Gravity (**accept any in range**)
A	Almandine	Garnet	Vitreous	3.93-4.30 g/cm^3
B	Chrysocolla	Silicates	Vitreous, glassy, greasy, earthy	1.90-2.45 g/cm^3

1. Almandine A
2. Chrysocolla B
3. Almandine A
4. Primarily through medium-to-high grade regional metamorphism of iron-rich argillaceous sediments and volcanic rocks

Station 9

Letter	Name and Type	Formula
A	Orthoclase/Microcline, Mineral	$KAlSi_3O_8$
B	Granite, Rock	Silica/SiO_2

1. Feldspar
2. Pink/white rectangular crystals
3. Na, Ca, K
4. Albite
5. Architecture, gravestones, memorials, engineering, paving- many options

Station 10

Letter	A	B	C	D	E
ID	Muscovite	Biotite	Olivine	Talc	Rock Crystal
Silicate Struct **(Div C Only)**	Phyllosilicate/ sheet	Phyllosilicate/ sheet	Nesosilicate/ Isolated tetrahedra	Phyllosilicate/ sheet	Tectosilicate/ Framework

1. Talc D- weak bonding between sheets
2. Olivine C- First mineral in Bowen's series.
3. Biotite B-
4. Biaxial
5. Massive to platy

Station 11

Letter	ID	Luster (accept any listed)	Specific Gravity **(accept anything within range)**
A	Pumice	Silky, dull, earthy, vitreous	2.0 - 2.85 g/cm^3
B	Obsidian	Vitreous, glassy, shiny	2.35 - 2.60 g/cm^3

1. Exsolving gases form bubbles during rapid cooling
2. Obsidian B- its thick, polymerized nature prevents atoms from organizing into crystals during rapid cooling
3. Pumice A
4. Pumice A- high silica content, rapid gas expansion, high porosity

Station 12

1.
    a. Ripple marks
    b. Mud cracks
    c. Cross bedding
    d. Graded bedding
2.
    a. Low to moderate
    b. Very low
    c. Moderate to high
    d. High to decreasing
3. B (Mud cracks) - form when wet sediment dries and shrinks after exposure to air
4. C (Cross bedding) - inclined layers dip in the direction of sediment transport.
5. Accept any reasonable explanation
6. B < A < C < D

Station 13

1.
    a. Cooling
    b. Melting
    c. Weathering & lithification
    d. Igneous rock
    e. Magma
    f. Melting
    g. Heat & pressure
    h. Weathering & lithification
    i. Sediment
    j. Compaction & cementation
    k. Heat & pressure
    l. Metamorphic rock
    m. Sedimentary rock
2. Weathering and erosion — climate controls temperature, precipitation, and biological activity, which affect weathering rates and sediment transport.- Other answers can get partial credit
3. Not a fixed sequence, multiple possible pathways, steps can be skipped or repeated, driven by internal (Earth) and external (Sun) energy- only need 2
4. Little/no subduction, reduced metamorphism and magma production, fewer volcanoes and mountain belts, dominated by surface processes (weathering, erosion, deposition)- only need 2
5.
    a. Burial- slate, heat- schist or gneiss, partial melting- magma, granite, or rhyolite

    b. Burial- low grade metamorphism, heat- regional metamorphism, partial melting- melting & crystallization

Station 14

1. MRAEYPGF$_a$SBWF$_B$T

2. Angular unconformity

3. Superposition

4. Principle of cross cutting relationship

5. Fossils occur in a predictable order, so comparing fossil types in each layer shows which layer is older or younger.

6. Relative dating only shows age order, while absolute dating gives a numerical age using radiometric methods.

Station 15

1. Rhyolite
2. Setting: Continental volcanic arcs or continental hotspots
   Why explosive: High silica → high viscosity → traps gases
3. Phyllite
4. Directed pressure during regional metamorphism, causing mineral alignment
5. Fossiliferous limestone
6. Accumulation and lithification of shell material from marine organisms
7. Peridotite
8. Obduction of mantle material (or emplacement at ophiolite complexes)
9. Greywacke
10. Preserves graded bedding from turbidity currents, indicating submarine slope processes and flow direction.

Station 16

1.

Beryl	Azurite	Augite	Amethyst
Chalcopyrite	Selenite	Celestite	Topaz
Sodalite	Citrine	Hornblende	Fluorite

2.

Slate	Coquina	Scoria	Dolostone
Pegmatite	Peridotite	Lignite	Arkose

BullSO, University of South Florida - Repaired practice edition - Answer key

Division C | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (2 points) Authored written substitute for unavailable source material: The best identification uses…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. Only specimen size

B. Several diagnostic observations

C. Color alone

D. One nickname

Correct: B